

Antibiotics in Microbiology

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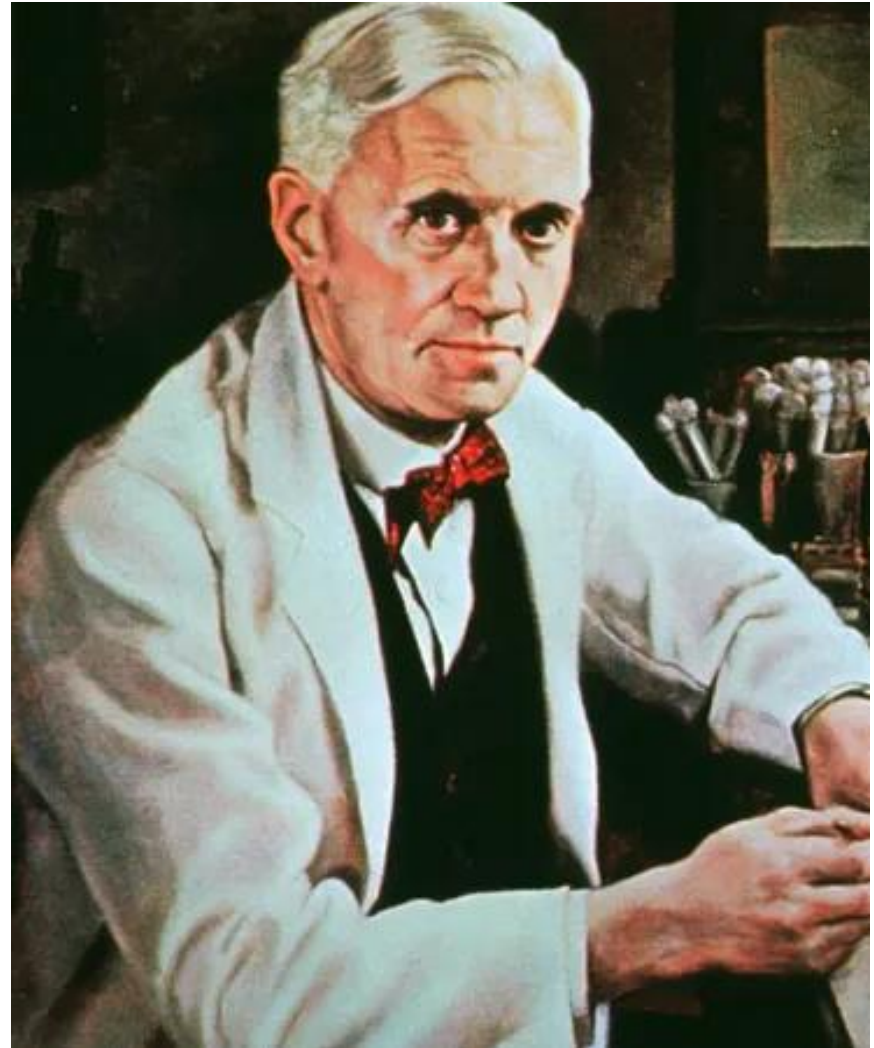
Lecture Objectives

- By the end of this lecture, students should be able to:
 - Define antibiotics and their role in treating infections.
 - Classify antibiotics based on structure and mechanism.
 - Understand bacterial resistance mechanisms.
 - Recognize clinical uses and adverse effects.
 - Apply knowledge in rational antibiotic prescribing.



Introduction to Antibiotics

- Antibiotics are natural, semi-synthetic, or synthetic compounds used to inhibit or kill microorganisms. They revolutionized modern medicine by reducing mortality from infectious diseases. Discovered by Alexander Fleming in 1928, penicillin marked the start of the antibiotic era. Understanding their types, actions, and resistance patterns is essential for clinical practice.



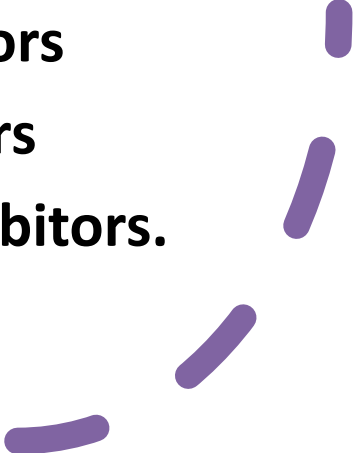
Classification of Antibiotics (1)

Antibiotics can be classified based on:

1. Spectrum of activity:

- **Broad-spectrum: act on Gram+ & Gram- bacteria.**
- **Narrow-spectrum: target specific bacteria.**

2. Mechanism of action:

- **Cell wall synthesis inhibitors**
 - **Protein synthesis inhibitors**
 - **Nucleic acid synthesis inhibitors.**
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Examples by class:

- **Beta-lactams: penicillins, cephalosporins.**
- **Aminoglycosides: gentamicin, amikacin.**
- **Macrolides: erythromycin, azithromycin.**
- **Tetracyclines: doxycycline.**
- **Fluoroquinolones: ciprofloxacin.**
- **Glycopeptides: vancomycin.**



Proper classification aids in selecting the right antibiotic for specific infections.



Mechanism of Action

1. Inhibition of cell wall synthesis:

- Beta-lactams bind to PBPs, preventing peptidoglycan formation.
- Example: penicillin, cephalosporins.

2. Inhibition of protein synthesis:

- Bind to 30S or 50S ribosomal subunits.
- Example: tetracyclines, aminoglycosides, macrolides.



3. Inhibition of nucleic acid synthesis:

- **Fluoroquinolones inhibit DNA gyrase.**
- **Rifampicin blocks RNA polymerase.**

4. Disruption of cell membrane function:

- **Polymyxins cause leakage of cell contents.**

Understanding mechanisms helps avoid unnecessary antibiotic use and resistance.



Spectrum of Activity

Antibiotics can be:

- **Broad-spectrum:** act against many Gram+ & Gram- bacteria, e.g., tetracyclines, carbapenems.
- **Narrow-spectrum:** effective against specific organisms, e.g., penicillin G for *Streptococcus*.

Clinical choice depends on suspected organism, site of infection, and resistance patterns.



Pharmacokinetics Overview

Key factors influencing antibiotic efficacy:

- **Absorption: oral vs IV bioavailability.**
- **Distribution: tissue penetration and protein binding.**
- **Metabolism: liver vs renal clearance.**
- **Excretion: urine or bile pathways.**

Understanding pharmacokinetics optimizes dosing and minimizes toxicity.

Antibiotic Resistance

Resistance occurs when bacteria adapt to survive antibiotic exposure. Types:

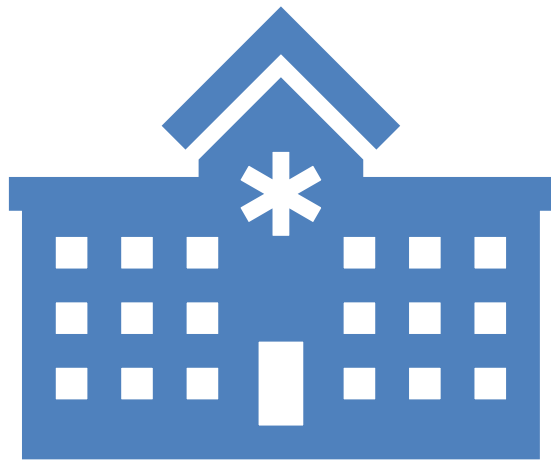
- Intrinsic: natural resistance due to structural traits.**
- Acquired: via mutations or horizontal gene transfer.**

Resistance leads to treatment failures, requiring rational prescribing and infection control.

Mechanisms include:

- **Enzymatic degradation (e.g., beta-lactamases).**
- **Altered target sites (e.g., MRSA modifies PBP).**
- **Efflux pumps removing drug from the cell.**
- **Reduced permeability of bacterial cell walls.**





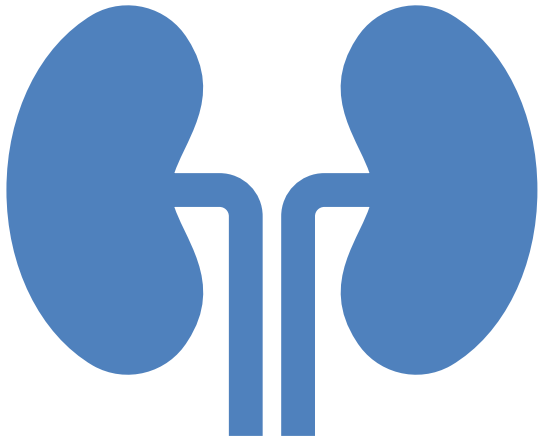
Clinical impact:

- **Longer hospital stays.**
- **Increased mortality.**
- **Higher treatment costs.**

Strategies to combat resistance:

- **Antibiotic stewardship.**
- **Infection prevention.**
- **Development of new agents.**

Clinical Uses of Antibiotics



Common indications:

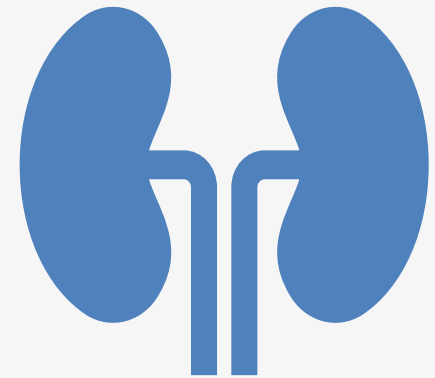
- **Upper and lower respiratory infections.**
- **Urinary tract infections.**
- **Skin and soft tissue infections.**

Choice depends on likely pathogens, severity, and local guidelines.

Examples of Clinical Uses of Antibiotics

- **Penicillins: streptococcal pharyngitis.**
- **Cephalosporins: pneumonia.**
- **Aminoglycosides: sepsis.**
- **Fluoroquinolones: complicated UTIs.**

Always consider patient allergies and comorbidities.



Side Effects & Toxicities

Common side effects:

- Gastrointestinal: nausea, diarrhea.
- Hypersensitivity reactions: rash, anaphylaxis.
- Organ-specific toxicity:
 - Aminoglycosides → nephrotoxicity.
 - Tetracyclines → hepatotoxicity.



Special considerations:

- **Avoid tetracyclines in children (tooth discoloration).**
- **Fluoroquinolones may cause tendon rupture.**
- **Macrolides can prolong QT interval.**

Close monitoring prevents severe complications.



Rational Use of Antibiotics

Principles:

- Confirm infection before prescribing.
- Use narrow-spectrum agents when possible.
- Adhere to recommended dosages and duration.
- Avoid unnecessary broad-spectrum use.

Goal: optimize outcomes, minimize resistance, reduce side effects.

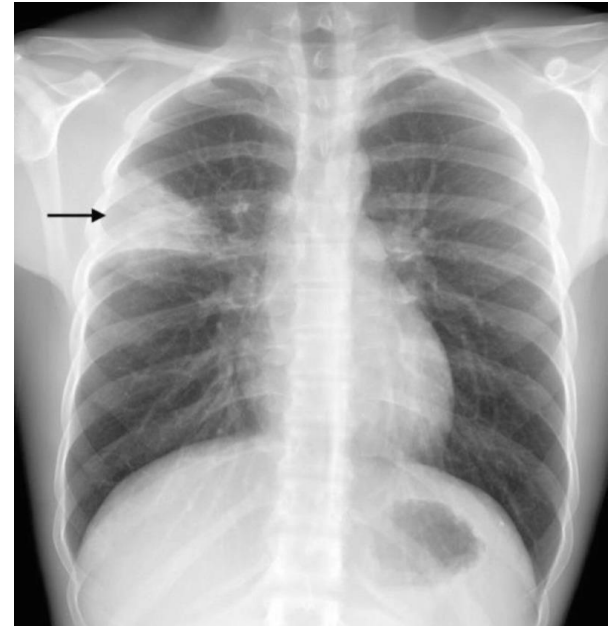


Case-Based Example

Case: A 35-year-old presents with fever, cough, and chest X-ray showing lobar consolidation.

- Likely diagnosis: community-acquired pneumonia.
- Empirical treatment: amoxicillin-clavulanate or macrolide.

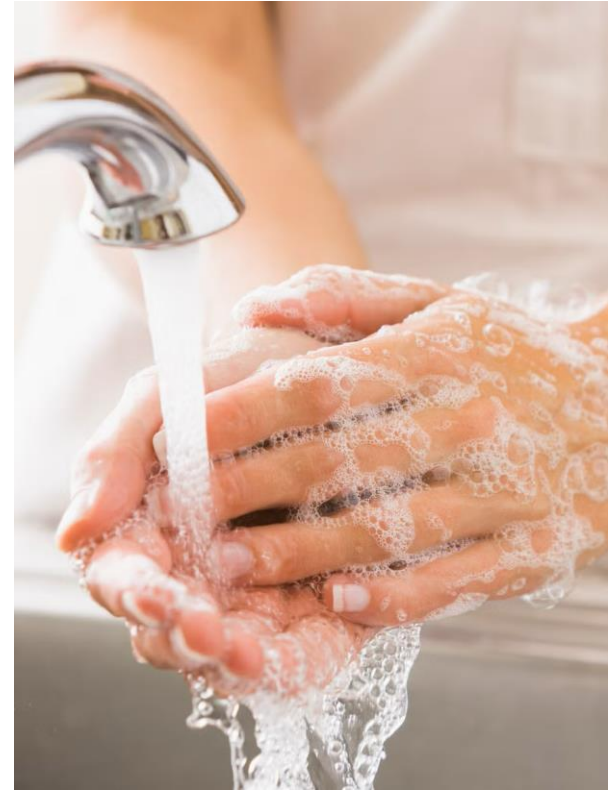
Culture results refine therapy choices.



Prevention of Resistance

Key measures:

- Hand hygiene and infection control.
- Vaccination to reduce antibiotic demand.
- Education on misuse and self-medication.
- Hospital antibiotic stewardship programs.



Summary

- **Antibiotics are powerful tools against infections but must be used responsibly.**
- **Understanding their classification, mechanisms, resistance, and clinical uses is vital.**
- **Rational prescribing protects patients and preserves antibiotic effectiveness.**



References

- 1. Jawetz, Melnick & Adelberg's Medical Microbiology, 28th Edition.
- 2. Murray PR, Rosenthal KS, Pfaller MA. Medical Microbiology, 9th Edition.
- 3. CDC Antibiotic Resistance Threats Report, 2024.